# The `tikz` package

This is a general purpose graphics package.[1] To load it for this document, I used:

```
\usepackage{tikz}
\usetikzlibrary{matrix,arrows,decorations.pathmorphing}
```

With `tikz` you can probably produce any diagram you need with any precision you desire, but the code can be complex. You may not wish to use it for a simple diagram:

$$\begin{tikzcd} A \arrow[r, "a"] \arrow[d, "b"] & B \arrow[d, "c"] \\ C \arrow[r, "d"] & D \end{tikzcd}$$

```
\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=2.6em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{A&B\\
C&D\\};
\path[->,font=\scriptsize,>=angle 90]
(m-1-1) edge node[auto] {$a$} (m-1-2)
        edge node[auto] {$b$} (m-2-1)
(m-1-2) edge node[auto] {$c$} (m-2-2)
(m-2-1) edge node[auto] {$d$} (m-2-2);
\end{tikzpicture}
```

$$\begin{CD} A @>a>> B \\ @VVbV @VVcV \\ C @>d>> D \end{CD}$$

```
$\begin{CD}
A@>a>>B\\
@VVbV@VVcV\\
C@>d>>D
\end{CD}$
(amscd)
```

This code sets up a matrix named `m` with some options, and then places $A$, $B$, $C$, and $D$ at the four positions of a $2 \times 2$ matrix. The next line specifies normal arrows with labels in scriptsize and a nondefault arrow head, and the following line specifies an arrow from the (1,1) position of the matrix `m` to the (1,2) position with a label $a$ in the default position.

However, `tikz` handles large objects and tall labels better than amscd:

$$\begin{tikzcd} A \times A \times A \times A \times A \times A \arrow[r, "a"] \arrow[d, "b"] & B \arrow[d, "c"] \\ C \arrow[r, "d"] & D \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r, "\overline{A^{A^A}}"] & D \end{tikzcd}$$

To my eyes, the arrow heads are too small. This can be fixed by adding `>=angle 90`, as an option to the path or to the whole picture:

$\longrightarrow$ `\path[->](1,1) edge (2,1);`

$\longrightarrow$ `\path[->,>=angle 90](1,1) edge (2,1);`

---

This is part of: Guide to Commutative Diagrams, `www.jmilne.org/not/CDGuide.html`
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[1] Felix Lenders has posted a tutorial "Commutative Diagrams using TikZ" at `http://www.felixl.de/commu.pdf`.

Here is the code for some arrows.

$\longrightarrow$ `\path[->](1,1) edge (2,1);`

$\longmapsto$ `\path[|->](1,1) edge (2,1);`

$\overline{\phantom{xxxx}}$ `\path[-](1,1) edge (2,1);`

$\hookrightarrow$ `\path[right hook->](1,1) edge (2,1);`

$\twoheadrightarrow$ `\path[->>](1,1) edge (2,1);`

$\cdots\!\rightarrow$ `\path[dotted,->](1,1) edge (2,1);`

$\dashrightarrow$ `\path[dashed,->](1,1) edge (2,1);`

$\bullet\!\longrightarrow$ `\path[*->](1,1) edge (2,1);`

$\rightsquigarrow$ `http://tex.stackexchange.com/questions/12678/`

The next example illustrates the use of the different arrows in a commutative diagram

$$\begin{tikzcd} A \arrow[r, hook] \arrow[dr, two heads] & B \arrow[d, dotted] & C \arrow[l, hook'] \arrow[dl, two heads] \\ & D & \end{tikzcd}$$

```
\begin{tikzpicture}[>=angle 90]
\matrix(a)[matrix of math nodes,
row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\
&D\\};
\path[right hook->](a-1-1) edge (a-1-2);
\path[->>](a-1-1) edge (a-2-2);
\path[dotted,->](a-1-2) edge (a-2-2);
\path[left hook->](a-1-3) edge (a-1-2);
\path[->>](a-1-3) edge (a-2-2);
\end{tikzpicture}
```

Now an example with labels on the arrows:

$$\begin{tikzcd} A \arrow[r, hook, "u", "b"'] \arrow[dr, two heads, "u", "b"'] & B \arrow[d, dotted, "u"', "b"] & C \arrow[l, hook', "u"', "b"] \arrow[dl, two heads, "u"', "b"] \\ & D & \end{tikzcd}$$

```
\begin{tikzpicture}[>=angle 90]
\matrix(a)[matrix of math nodes, row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\&D\\};
\path[right hook->,font=\scriptsize]
(a-1-1) edge node[above]{$u$} node[below]{$b$} (a-1-2);
\path[->>,font=\scriptsize]
(a-1-1) edge node[above]{$u$} node[below]{$b$} (a-2-2)
(a-1-3) edge node[above left]{$u$} node[below right]{$b$} (a-2-2);
\path[dotted,->,font=\scriptsize]
(a-1-2) edge node[left]{$u$} node[right]{$b$} (a-2-2);
\path[left hook->,font=\scriptsize]
```

```
(a-1-3) edge node[above]{$u$} node[below]{$b$} (a-1-2);
\end{tikzpicture}
```

Long labels may cause a problem:

$$\begin{tikzcd}
A \arrow[r] \arrow[d] & B \arrow[r, "\text{very long label}"] \arrow[d] & C \arrow[d] \\
D \arrow[r] & E \arrow[r] & F
\end{tikzcd}$$

```
\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=3em, column sep=2.5em,
text height=1.5ex, text depth=0.25ex]
{A&B&C\\
D&E&F\\};
\path[->,font=\scriptsize]
(m-1-1) edge (m-1-2)
        edge (m-2-1)
(m-1-2) edge node[auto] {very long label} (m-1-3)
        edge (m-2-2)
(m-1-3) edge (m-2-3)
(m-2-1) edge (m-2-2)
(m-2-2) edge (m-2-3);
\end{tikzpicture}
```

However, this can be fixed by setting `column sep=5.0em`.

$$\begin{tikzcd}[column sep=5.0em]
A \arrow[r] \arrow[d] & B \arrow[r, "\text{very long label}"] \arrow[d] & C \arrow[d] \\
D \arrow[r] & E \arrow[r] & F
\end{tikzcd}$$

Nor does it have a problem with objects of different heights.

$$\begin{tikzcd}
\hat{A} \arrow[r] & \prod_{n\in Z} A_n \arrow[r] & \prod_{n\in Z} A_n.
\end{tikzcd}$$

But that is because of the options `text height=1.5ex, text depth=0.25ex`. When you omit them, you get:

$$\begin{tikzcd}
\hat{A} \arrow[r] & \prod_{n\in Z} A_n \arrow[r] & \prod_{n\in Z} A_n.
\end{tikzcd}$$

Curving arrows is easy.

$$
\begin{tikzcd}
A \arrow[r, bend left] \arrow[r, bend left=40] \arrow[r, bend left=60] \arrow[r, bend left=80] \arrow[r, bend right] & B
\end{tikzcd}
$$

```
\begin{tikzpicture}
\matrix(m)[matrix of math nodes,
row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{A&B\\};
\path[->]
(m-1-1) edge [bend left] (m-1-2)
        edge [bend left=40] (m-1-2)
        edge [bend left=60] (m-1-2)
        edge [bend left=80] (m-1-2)
        edge [bend right] (m-1-2);
\end{tikzpicture}
```

Two more examples:

$$
\begin{tikzcd}
& A \arrow[dl, "f"'] \arrow[dr, "c" description] \arrow[drr, "d" description] \arrow[drrr, "e" description] \\
& B \arrow[dl] \arrow[r] & C \arrow[r] & D \arrow[r] & E \\
F
\end{tikzcd}
$$

```
\[
\begin{tikzpicture}[descr/.style={fill=white}]
\matrix(m)[matrix of math nodes, row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{&A\\&B&C&D&E\\F\\};
\path[->,font=\scriptsize]
(m-1-2) edge node[above left] {$f$} (m-3-1)
        edge node[descr] {$c$} (m-2-3)
        edge node[descr] {$d$} (m-2-4)
        edge node[descr] {$e$} (m-2-5);
\path[->]
(m-2-2) edge (m-3-1)
        edge (m-2-3);
\path[->]
(m-2-3) edge (m-2-4);
\path[->]
(m-2-4) edge (m-2-5);
\end{tikzpicture}
\]
```

$$
\begin{tikzcd}
T \arrow[drr, bend left, "x"] \arrow[ddr, bend right, "y"'] \arrow[dr, dotted, "{(x,y)}" description] & & \\
 & X \times_Z Y \arrow[r, "p"'] \arrow[d, "q"] & X \arrow[d, "f"] \\
 & Y \arrow[r, "g"] & Z
\end{tikzcd}
$$

```
\[
\begin{tikzpicture}[descr/.style={fill=white}]
\matrix(m)[matrix of math nodes, row sep=3em, column sep=2.8em,
text height=1.5ex, text depth=0.25ex]
{T\\&X\times_Z Y&X\\&Y&Z\\};
\path[->,font=\scriptsize]
(m-1-1) edge [bend left=10] node[above] {$x$} (m-2-3)
(m-1-1) edge [bend right=10] node[below] {$y$} (m-3-2);
\path[->,dotted,font=\scriptsize]
(m-1-1) edge node[descr] {$(x,y)$} (m-2-2);
\path[->,font=\scriptsize]
(m-2-2) edge node[below] {$p$} (m-2-3)
(m-2-2) edge node[right] {$q$} (m-3-2);
\path[->,font=\scriptsize]
(m-2-3) edge node[right] {$f$} (m-3-3);
\path[->,font=\scriptsize]
(m-3-2) edge node[above] {$g$} (m-3-3);
\end{tikzpicture}
\]
```

When you number a displayed commutative diagram

```
\begin{equation}
\begin{tikzpicture}
......
\end{tikzpicture}
\end{equation}
```

$$
\begin{tikzcd}
A \arrow[r, "a"] \arrow[d, "c"] & B \arrow[r, "b"] \arrow[d, "d"] & C \arrow[d, "e"] \\
D \arrow[r, "f"] & E \arrow[r, "g"] & F
\end{tikzcd}
\tag{1}
$$

the number appears below the level of the diagram. To centre the number, use:

```
\begin{equation}
\begin{tikzpicture}[baseline=(current bounding box.center)]
......
\end{tikzpicture}
\end{equation}
```

$$
\begin{tikzcd}
A \arrow[r, "a"] \arrow[d, "c"] & B \arrow[r, "b"] \arrow[d, "d"] & C \arrow[d, "e"] \\
D \arrow[r, "f"] & E \arrow[r, "g"] & F
\end{tikzcd}
\tag{2}
$$

After a while, `tikz` becomes addictive: The extended snake lemma says that the exact commutative diagram in blue gives rise to the exact sequence in red.

$$
\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker} f \arrow[r] & \operatorname{Ker} a \arrow[r] \arrow[d] & \operatorname{Ker} b \arrow[r] \arrow[d] & \operatorname{Ker} c \arrow[d] \arrow[dddll, "d"'] & & \\
& & A \arrow[r, "f"] \arrow[d, "a"] & B \arrow[r] \arrow[d, "b"] & C \arrow[r] \arrow[d, "c"] & 0 & \\
& 0 \arrow[r] & A' \arrow[r] \arrow[d] & B' \arrow[r, "g'"] \arrow[d] & C' \arrow[d] & & \\
& & \operatorname{Coker} a \arrow[r] & \operatorname{Coker} b \arrow[r] & \operatorname{Coker} c \arrow[r] & \operatorname{Coker} g' \arrow[r] & 0
\end{tikzcd}
$$

```
\begin{tikzpicture}[>=angle 90]
\matrix[matrix of math nodes,
row sep=3em, column sep=3em,,
text height=1.5ex, text depth=0.25ex]
{|[name=00]|0&|[name=kf]|\Ker f&|[name=ka]| \Ker a &|[name=kb]|
\Ker b &|[name=kc]| \Ker c \\
&&|[name=A]| A &|[name=B]| B &|[name=C]| C &|[name=01]| 0 \\
&|[name=02]| 0 &|[name=A']| A' &|[name=B']| B' &|[name=C']| C' \\
&&|[name=ca]| \Coker a &|[name=cb]| \Coker b &|[name=cc]|
\Coker c &|[name=cg]|\Coker g'&|[name=04]|0\\};
\draw[->,font=\scriptsize]
          (ka) edge (A)
          (kb) edge (B)
          (kc) edge (C);
\draw[->,font=\scriptsize,blue,thick]
          (A) edge node[auto] {$f$} (B)
          (B) edge (C)
          (C) edge (01)
          (A) edge node[auto] {$a$} (A')
          (B) edge node[auto] {$b$} (B')
          (C) edge node[auto] {$c$} (C')
```

```
          (02) edge (A')
          (A') edge (B')
          (B') edge node[auto] {$g'$} (C');
\draw[->,font=\scriptsize]
          (A') edge (ca)
          (B') edge (cb)
          (C') edge (cc);
\draw[->,red]
          (00) edge (kf)
          (kf) edge (ka)
          (ka) edge (kb)
          (kb) edge (kc)
          (ca) edge (cb)
          (cb) edge (cc)
          (cb) edge (cc)
          (cc) edge (cg)
          (cg) edge (04)
          (kc) edge[out=0,in=180,red] node[above left] {$d$} (ca);
\end{tikzpicture}
```

For the last diagram, I added the following lines to the preamble

```
\usepackage{amsmath}
\DeclareMathOperator{\Coker}{Coker}
\DeclareMathOperator{\Ker}{Ker}
```